Lesson 9.4 - Finding a Confidence Interval of a Population Proportion

> **Video:** Finding The Confidence Interval of a Population Proportion Using The Normal Distribution by The Organic Chemistry Tutor **Chapter:** 9 - Confidence Intervals

The big idea

You want to know what **fraction** of a large group has some property, for example "what percent of students at this university own a laptop?" Asking everyone (maybe 10,000 people) is too much work, so you ask a **sample** instead.

The sample gives you **one estimate**. Because it's only a sample, that estimate is probably a little off. A confidence interval turns the single estimate into a **range** and says how confident you are that the true value is inside it:

$$\underbrace{\hat{p}}_{\text{your estimate}} \pm \underbrace{E}_{\text{margin of error}}$$

Earlier lessons did this for a **mean** (an average of numbers). This lesson does the same thing for a **proportion** (a percentage of yes answers).

The symbols

The letters are the most confusing part of this lesson, so here they all are in one place:

Symbol	Name	Meaning	Laptop example
n	sample size	how many people you asked	800
x	number of successes	how many said **yes**	584
p	population proportion	the **true** fraction of everyone. Unknown, and it's what we're trying to estimate	?
$\hat{p}$ ("p-hat"), also written p' ("p-prime")	sample proportion	the fraction of **your sample** that said yes: $\hat{p} = x/n$	$584/800 = 0.73$
$\hat{q}$ ("q-hat")	proportion of failures	the fraction of your sample that said **no**: $\hat{q} = 1 - \hat{p}$	0.27
CL	confidence level	how sure you want to be, as a decimal	95% → 0.95
z	critical value	how many standard errors wide the interval is; depends only on CL	1.96
E or EBP	margin of error, or "error bound for a proportion"	how far the interval reaches on each side of $\hat{p}$	0.0308

> **p vs $\hat{p}$:** p (no hat) is the true answer for the whole population, which you never actually know. $\hat{p}$ (with hat) is what your sample gave you. The confidence interval is a range of believable values for p, built around $\hat{p}$.

$\hat{p}$ and $\hat{q}$ always add up to 1. If 73% said yes, then 27% said no.

Formulas

Quantity	Formula
Sample proportion	$\hat{p} = \frac{x}{n}$
Proportion of failures	$\hat{q} = 1 - \hat{p}$
Check: is the normal distribution OK?	$n\hat{p}\hat{q} \geq 10$
Area to the left of z	$\frac{1 + \text{CL}}{2}$
Margin of error	$E = z\sqrt{\frac{\hat{p}\,\hat{q}}{n}}$
Confidence interval	$\hat{p} - E < p < \hat{p} + E$
Sample size needed	$n = \frac{z^2\,\hat{p}\,\hat{q}}{E^2}$, always rounded **up**

Common z values:

Confidence level	Area to the left	z
90%	0.95	1.645
95%	0.975	1.96
98%	0.99	2.33
99%	0.995	2.575

The recipe (do these steps in order)

1. **Write down n and x** from the problem.
2. **Find $\hat{p} = x/n$ and $\hat{q} = 1 - \hat{p}$.**
3. **Check $n\hat{p}\hat{q} \geq 10$.** If it is, the normal distribution is a good approximation and you can use z.
4. **Find z** from the confidence level (see the next section).
5. **Find the margin of error $E = z\sqrt{\hat{p}\hat{q}/n}$.**
6. **Build the interval:** lower end $\hat{p} - E$, upper end $\hat{p} + E$.
7. **Write a sentence** saying what it means.

Step 3 in detail - why check $n\hat{p}\hat{q} \geq 10$?

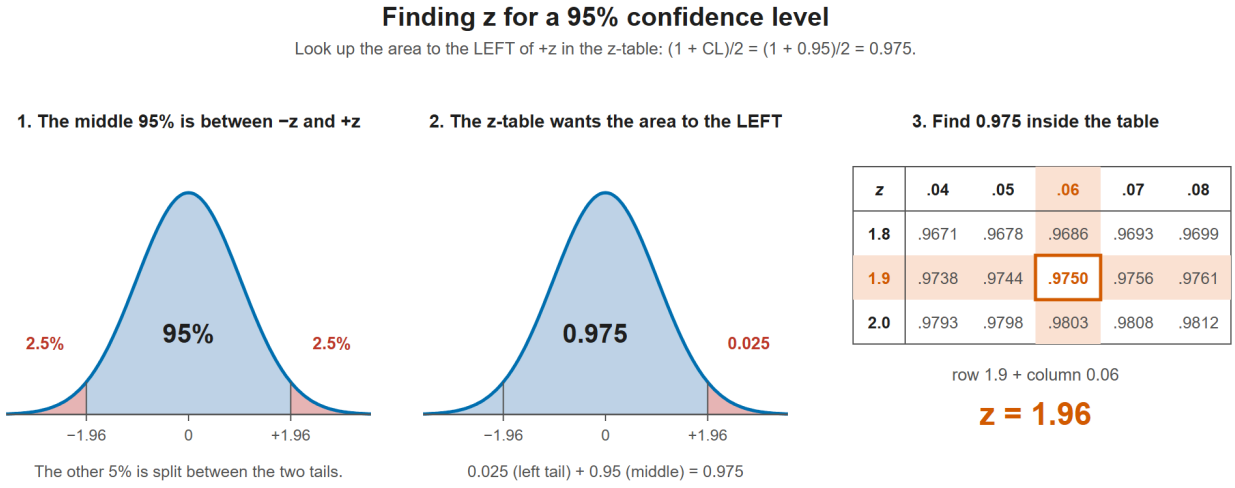
Each person gives one of two answers (yes or no), so the number of yeses follows a **binomial distribution**. The confidence interval formula uses the **normal** (bell) curve instead, because z values come from the normal curve.

That's only allowed when the binomial distribution looks enough like a bell curve, which happens when the sample is big enough and $\hat{p}$ isn't too close to 0 or 1. The rule of thumb $n\hat{p}\hat{q} \geq 10$ tests this.

> Many textbooks use a slightly different check: $n\hat{p} \geq 10$ **and** $n\hat{q} \geq 10$, meaning at least 10 yeses and at least 10 noes. Use whichever your course uses.

Step 4 in detail - finding z from the confidence level

The picture. A 95% confidence level means the middle 95% of the bell curve, between $-z$ and $+z$. The other 5% is split equally between the two tails: 2.5% on the left and 2.5% on the right.



The three steps for 95%. Left: the confidence level is the middle area, and the leftover 5% is split into two 2.5% tails. Middle: the z-table measures from the far left, so the area it wants also includes the left tail, 0.975 in total. Right: 0.9750 sits in row 1.9 and column 0.06, so z = 1.96.

A z-table gives the area to the **left** of a z-score. The area to the left of $+z$ is the middle 95% plus the left tail 2.5%:

$$\text{CL} + \frac{1 - \text{CL}}{2} = 0.95 + 0.025 = 0.975$$

The shortcut formula gives the same answer without the picture:

$$\text{area to the left} = \frac{1 + \text{CL}}{2} = \frac{1 + 0.95}{2} = \frac{1.95}{2} = 0.975$$

Looking it up in the table. Find 0.975 **inside** the positive z-table, then read off its row and column:

- Row: 1.9
- Column: 0.06
- So $z = 1.9 + 0.06 = 1.96$

If the exact area isn't in the table, take the two closest entries and average their z-scores. For example, for 90% the area to the left is 0.95. The table has 0.9495 (at $z = 1.64$) and 0.9505 (at $z = 1.65$). 0.95 is exactly halfway, so

$$z = \frac{1.64 + 1.65}{2} = 1.645$$

Step 5 in detail - what the margin of error formula means

$$E = z \times \underbrace{\sqrt{\frac{\hat{p}\,\hat{q}}{n}}}_{\text{standard error}}$$

- The square-root part is the **standard error**: roughly how much $\hat{p}$ would vary if you repeated the survey with different random people.
- z sets how many standard errors you allow on each side. More confidence means a bigger z and a wider interval.
- A bigger n makes the standard error smaller, because larger samples are more reliable. That gives a narrower interval.

For why $\hat{p}\hat{q}$ appears here, see "The logic behind the formulas" in Lesson 9.5.

Example 1 - laptop ownership

800 randomly selected college students at university XYZ were asked if they own a laptop. 584 said yes.

(a) Can the binomial distribution be approximated by a normal distribution?

Write down what you know. $n = 800$ and $x = 584$. The true proportion p (all students at the university) is unknown, so we estimate it with $\hat{p}$.

Find $\hat{p}$ and $\hat{q}$.

$$\hat{p} = \frac{x}{n} = \frac{584}{800} = 0.73 \qquad \hat{q} = 1 - \hat{p} = 1 - 0.73 = 0.27$$

So 73% of the sample said yes and 27% said no.

Check the condition.

$$n\hat{p}\hat{q} = 800(0.73)(0.27) = 157.68 \geq 10 \checkmark$$

Yes, the normal distribution can be used.

(b) 95% confidence interval and margin of error

Find z. The area to the left is $(1 + 0.95)/2 = 0.975$, which gives $z = 1.96$.

Find the margin of error. Work from the inside out:

$$\frac{\hat{p}\,\hat{q}}{n} = \frac{0.73 \times 0.27}{800} = \frac{0.1971}{800} = 2.46375 \times 10^{-4}$$

$$\sqrt{\frac{\hat{p}\,\hat{q}}{n}} = \sqrt{2.46375 \times 10^{-4}} = 0.015696$$

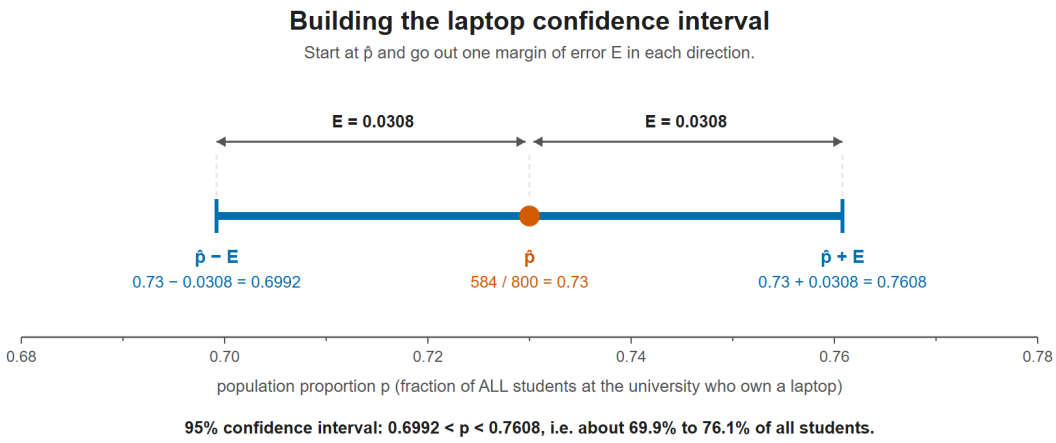
$$E = z\sqrt{\frac{\hat{p}\,\hat{q}}{n}} = 1.96 \times 0.015696 = 0.0307648 \approx 0.0308$$

Build the interval.

$$\text{lower} = \hat{p} - E = 0.73 - 0.0308 = 0.6992$$

$$\text{upper} = \hat{p} + E = 0.73 + 0.0308 = 0.7608$$

$$\boxed{0.6992 < p < 0.7608}$$



The interval is built outward from the sample result. The dot is $\hat{p}$ = 0.73, and each arm is one margin of error, so the two ends are $\hat{p} - E = 0.6992$ and $\hat{p} + E = 0.7608$.

Interpretation. We are 95% confident that between about **69.9% and 76.1%** of **all** students at university XYZ own a laptop.

The margin of error is 0.0308, about **±3.1 percentage points**.

(c) What sample size gives a 2% margin of error at 90% confidence?

This time we're **not** building an interval. We're asking how many people we would need to survey.

What changes. $\hat{p} = 0.73$ and $\hat{q} = 0.27$ stay the same. The margin of error is now $E = 2\% = 0.02$, the confidence level is 90%, and n is the unknown.

Solve the margin of error formula for n.

$$E = z\sqrt{\frac{\hat{p}\hat{q}}{n}}$$

Square both sides. The square and the square root cancel:

$$E^2 = \frac{z^2\,\hat{p}\hat{q}}{n}$$

Multiply both sides by n:

$$nE^2 = z^2\,\hat{p}\hat{q}$$

Divide both sides by E^2:

$$n = \frac{z^2\,\hat{p}\,\hat{q}}{E^2}$$

Find z for 90%. The area to the left is $(1 + 0.90)/2 = 0.95$, which gives $z = 1.645$. This is the "average two table values" case from Step 4.

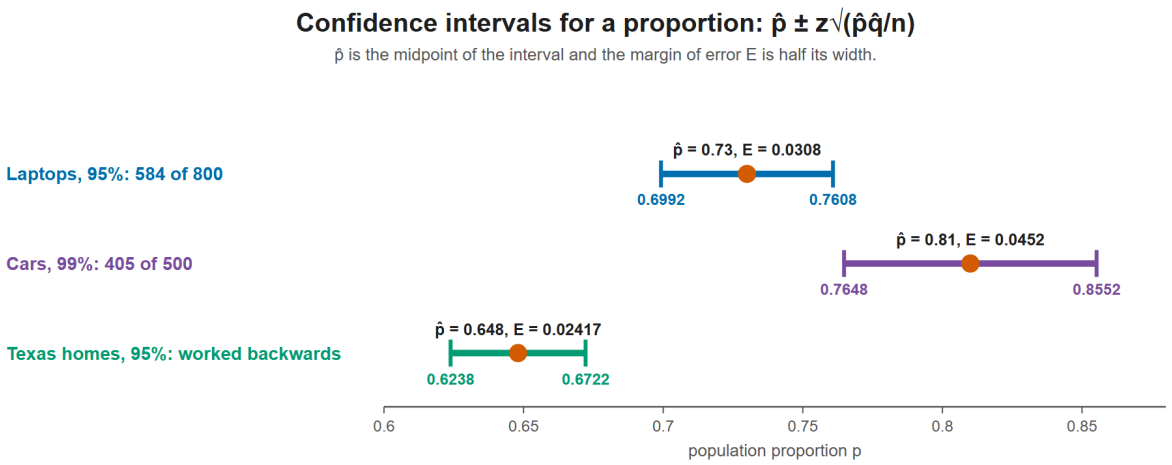
Plug in. Don't forget to square **both** z and E:

$$n = \frac{z^2\,\hat{p}\,\hat{q}}{E^2} = \frac{(1.645)^2(0.73)(0.27)}{(0.02)^2} = \frac{2.706025 \times 0.1971}{0.0004} = \frac{0.53336}{0.0004} \approx 1333.39$$

Round up, even though normal rounding would give 1333:

$$n = 1334 \text{ students}$$

> **Why always round up?** With 1333 students the margin of error would be slightly **more** than 2%. We need **at least** enough people to reach 2%, so we go up to the next whole person.



Texas: p̂ = (0.62383 + 0.67217)/2 = 0.648 and E = (0.67217 − 0.62383)/2 = 0.02417, then n = z²p̂q̂/E² ≈ 1,500.

The three intervals from the lesson. The dot is p̂, the midpoint, and each arm is the margin of error E. The 99% car interval is the widest, because higher confidence uses a larger z.

Example 2 - car ownership

500 residents in town ABC were asked if they own a car. 405 said yes.

Write down what you know and find $\hat{p}$, $\hat{q}$.

$$n = 500, \quad x = 405, \quad \hat{p} = \frac{x}{n} = \frac{405}{500} = 0.81, \quad \hat{q} = 1 - \hat{p} = 1 - 0.81 = 0.19$$

So 81% of the sample own a car.

(a) 99% confidence interval

Find z. The area to the left is $(1 + 0.99)/2 = 1.99/2 = 0.995$. In the table, 0.995 falls between $z = 2.57$ and $z = 2.58$, so average them:

$$z = \frac{2.57 + 2.58}{2} = 2.575$$

Find the margin of error.

$$\frac{\hat{p}\,\hat{q}}{n} = \frac{0.81 \times 0.19}{500} = \frac{0.1539}{500} = 0.0003078 \qquad \sqrt{\frac{\hat{p}\,\hat{q}}{n}} = \sqrt{0.0003078} = 0.01754$$

$$E = z\sqrt{\frac{\hat{p}\,\hat{q}}{n}} = 2.575 \times 0.01754 \approx 0.0452$$

Build the interval.

$$\hat{p} - E = 0.81 - 0.0452 = 0.7648 \qquad \hat{p} + E = 0.81 + 0.0452 = 0.8552$$

$$\boxed{0.7648 < p < 0.8552}$$

Interpretation. We are 99% confident that between about **76.5% and 85.5%** of all residents of town ABC own a car.

> Compare with Example 1: the margin of error here (±4.5 points) is wider than the laptop one (±3.1 points). Partly that's because 99% confidence uses a bigger z (2.575 vs 1.96), and partly because the sample is smaller (500 vs 800).

(b) What sample size gives a 3% margin of error at 98% confidence?

Find z. The area to the left is $(1 + 0.98)/2 = 1.98/2 = 0.99$. The closest table entry is 0.9901, at $z = 2.33$.

Plug in, with $E = 3\% = 0.03$:

$$n = \frac{z^2\,\hat{p}\,\hat{q}}{E^2} = \frac{(2.33)^2(0.81)(0.19)}{(0.03)^2} = \frac{5.4289 \times 0.1539}{0.0009} = \frac{0.83551}{0.0009} \approx 928.34$$

Round up: $n = 929$ residents.

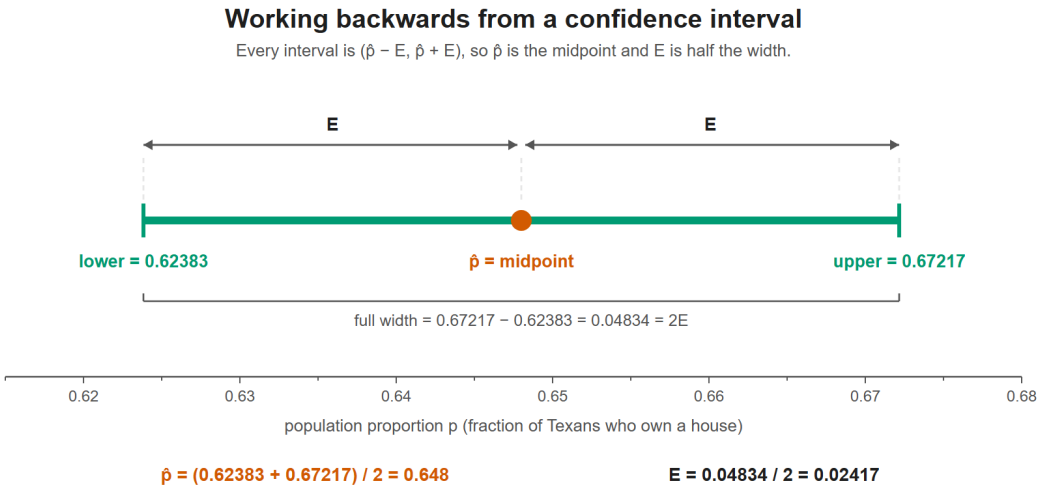
Example 3 - working backwards from an interval

Some number of people in Texas were asked if they own a house. The 95% confidence interval for the true proportion of homeowners was $(0.62383,\ 0.67217)$.

This time the interval is **given**, and we work back to find the pieces that built it. The key fact is that every confidence interval has the shape

$$(\hat{p} - E,\ \ \hat{p} + E)$$

so $\hat{p}$ sits **exactly in the middle**, and E is the distance from the middle to either end.



Only the two ends are given. Because the interval is p̂ ± E, p̂ is the point halfway between them (0.648), and E is half of the full width (0.04834 / 2 = 0.02417).

(a) What is the sample proportion?

$\hat{p}$ is the **midpoint**, the average of the two ends:

$$\hat{p} = \frac{\text{lower} + \text{upper}}{2} = \frac{0.62383 + 0.67217}{2} = \frac{1.296}{2} = 0.648$$

(b) What is the error bound?

Method 1: upper end minus the middle.

$$E = \text{upper} - \hat{p} = 0.67217 - 0.648 = 0.02417$$

Method 2: half the width of the interval. The full width is 2E, so

$$E = \frac{\text{upper} - \text{lower}}{2} = \frac{0.67217 - 0.62383}{2} = \frac{0.04834}{2} = 0.02417$$

Both methods give the same answer.

(c) How many people were surveyed?

That's n, so use the sample-size formula. At 95% confidence $z = 1.96$, and $\hat{q} = 1 - \hat{p} = 1 - 0.648 = 0.352$:

$$n = \frac{z^2\,\hat{p}\,\hat{q}}{E^2} = \frac{(1.96)^2(0.648)(0.352)}{(0.02417)^2} = \frac{3.8416 \times 0.228096}{0.000584189} = \frac{0.87625}{0.000584189} \approx 1499.95$$

So about **1500 people** were surveyed.

> Here 1499.95 is just rounding noise in the given interval, so the real answer is 1500. Rounding up and rounding to the nearest whole number both give 1500.

(d) How many said yes?

$\hat{p}$ is the **fraction** who said yes, and x is the **number** who said yes. Since $\hat{p} = x/n$, multiply both sides by n:

$$x = n\,\hat{p} = 1500 \times 0.648 = 972$$

So **972 people** said yes, which is 64.8% of 1500.

Common mistakes

- **Mixing up p and $\hat{p}$.** $\hat{p}$ comes from the sample, and you calculate it. p is the true population value, and you never know it exactly. The interval is a guess for p.
- **Forgetting to convert percentages.** A 2% margin of error is $E = 0.02$, not 2. A 95% confidence level is 0.95.
- **Looking up the confidence level directly in the z-table.** Look up the **area to the left**, $(1 + \text{CL})/2$. For 95%, look up 0.975, not 0.95.
- **Forgetting to square** z or E in the sample-size formula.
- **Rounding n down.** Sample sizes always round **up**.
- **Misreading what "95% confident" means.** It does **not** mean there's a 95% chance that p is in this particular interval. p is a fixed number, and it either is or isn't inside. It means the **method** works 95% of the time: if you repeated the survey many times and built an interval each time, about 95% of those intervals would contain the true p.

Key takeaways

- $\hat{p} = x/n$ is your estimate, and $\hat{q} = 1 - \hat{p}$. Check $n\hat{p}\hat{q} \geq 10$ before using the normal curve.
- Find z by looking up the area to the left, $(1 + \text{CL})/2$, in the z-table.
- Confidence interval: $\hat{p} \pm z\sqrt{\hat{p}\hat{q}/n}$.
- Higher confidence means a bigger z and a wider interval. A bigger sample means a narrower interval.

- Given an interval: its midpoint is $\hat{p}$, and half its width is E.
- Sample size: $n = z^2\hat{p}\hat{q}/E^2$, always rounded **up**. Lesson 9.5 covers sample size in more depth.